TechBazaar

E-commerce Sales Data Cleaning & Exploratory Analysis — Documentation

Python · pandas · matplotlib · seaborn | Source: Cleaning_TechBazaar.ipynb

1. Introduction

TechBazaar is an online retailer selling smartphones, tablets, laptops and accessories to customers across Indian cities. The raw exports from its systems are messy: city names, payment methods and product categories are spelled inconsistently, dates appear in three different formats, many values are missing and every table contains duplicate rows.

This document describes how the four raw tables were cleaned, merged into one analysis dataset (**1,624 rows × 22 columns**) and analysed to answer business questions on revenue, products, payments and cities. All work is in the notebook **Cleaning_TechBazaar.ipynb**, and every figure below comes from that notebook's outputs.

2. Objectives

- Clean and standardise the four raw CSV files (customers, orders, order items, products).
- Parse all order dates correctly, including the three mixed formats in the raw data.
- Remove customers with no city and orders with no payment method rather than guessing their values, and keep every table consistent with the rows that remain.
- Engineer time features (Month, Quarter, Year) and a line-item **Amount** = quantity × unit price.
- Analyse revenue by category, payment method, year, quarter, month and city.

3. Dataset Description

The data comes from four related CSV files. Customers link to orders through **customer_id**, orders link to order items through **order_id**, and order items link to products through **product_id**.

File	Rows	Columns	Missing values found
messy_customers.csv	**520**	customer_id, customer_name, city, signup_date	name **28**, city **41**
messy_orders.csv	**1,240**	order_id, customer_id, order_date, payment_method	payment_method **145**
messy_order_items.csv	**2,050**	order_item_id, order_id, product_id, quantity, unit_price	quantity **318**, unit_price **1,011**
messy_products.csv	**90**	product_id, product_name, category, base_price	base_price **57**

Table 1. Raw data files.

Table	Raw rows	Final rows	How rows were removed
Customers	520	**460**	**41** with no city, then **19** duplicates
Orders	1,240	**958**	**145** with no payment method, **36** duplicates, then **101** belonging to removed customers
Order items	2,050	**1,624**	**50** duplicates, then **376** belonging to removed orders
Products	90	**80**	**10** duplicates
Merged dataset	—	**1,624 × 22**	Customers + orders + order items + products

Table 2. Row counts before and after cleaning.

4. Data Cleaning & Transformation

Table	Issue found	Fix applied
Customers	Missing names and cities; city spelled Delhi / delhi / DELHI, Bengaluru vs Bangalore	Missing names set to **Unknown**; rows with **no city removed**; spellings standardised to **7 cities**; duplicates dropped; signup_date parsed (all ISO)
Orders	Missing payment method; UPI / upi, Credit Card / credit card, Cash / Cash on Delivery; **three date formats**	Rows with **no payment method removed**; labels mapped to **4 clean values**; duplicates dropped; dates parsed **explicitly per format** (see below)
Order items	Missing quantity and unit_price; zero quantities	Quantity filled with **1** and zeros set to 1; unit_price filled with the **product median** (overall median if none); duplicates dropped
Products	Missing base_price; categories like smart phone, Accessory, laptop	base_price filled with the mean (**≈ ₹45,047**); categories standardised to **4 clean values**; duplicates dropped

Table 3. Cleaning steps by table.

4.1 Date parsing

Order dates appear in three formats, so each is matched with its own pattern and parsed with an explicit format. This avoids the ambiguity of day-first versus month-first guessing.

Format	Example	Parsed as
Year-month-day (ISO)	2024-02-01	%Y-%m-%d
Day/month/year with slashes	13/04/2025	%d/%m/%Y
Month-day-year with dashes	04-26-2025	%m-%d-%Y

Table 4. Order date formats.

All **1,059** orders with a payment method parse successfully, and dates run from **1 January 2024** to **15 May 2025**.

4.2 Keeping the tables consistent

After customers with no city and orders with no payment method are removed, orders from removed customers are dropped, and then order items belonging to dropped orders are dropped. Because of this, **every result below is based on the same 1,624 order items** and adds up to the same total of **₹145,138,977.5**. Month, Quarter and Year are derived from order_date and signup_date, and **Amount** is calculated for every remaining item.

5. Insights

5.1 Revenue and top category

Total revenue is **₹145,138,977.5** (about **₹14.51 crore**) across **3,261** units. **Smartphone** leads on both volume (**1,033 units**) and revenue (**₹43.60M**, 30.0% of the total). Accessories rank second on both measures, and Laptop is last with **623 units** and **₹26.06M**.

Category	Units sold	Revenue	Share of revenue
Smartphone	**1,033**	**₹43.60M**	30.0%
Accessories	858	₹39.79M	27.4%
Tablet	747	₹35.68M	24.6%
Laptop	623	₹26.06M	18.0%
Total	**3,261**	**₹145.14M**	**100%**

Table 5. Units and revenue by category.

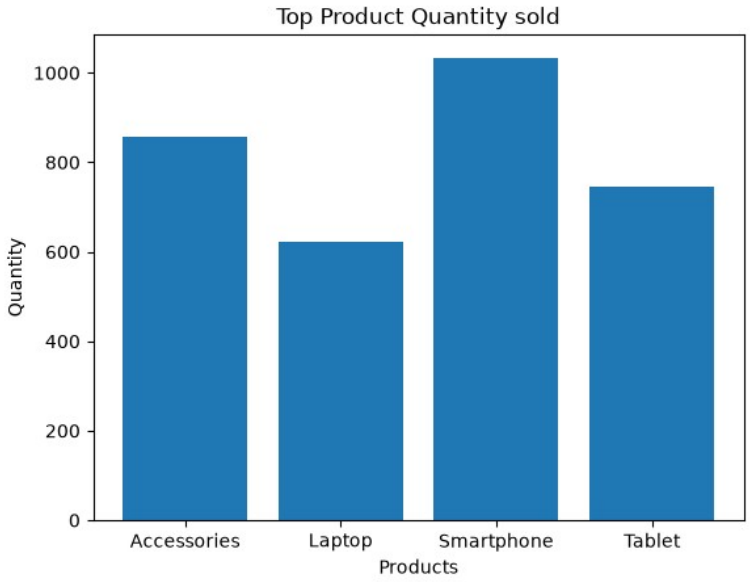


Figure 1. Units sold per category (matplotlib).

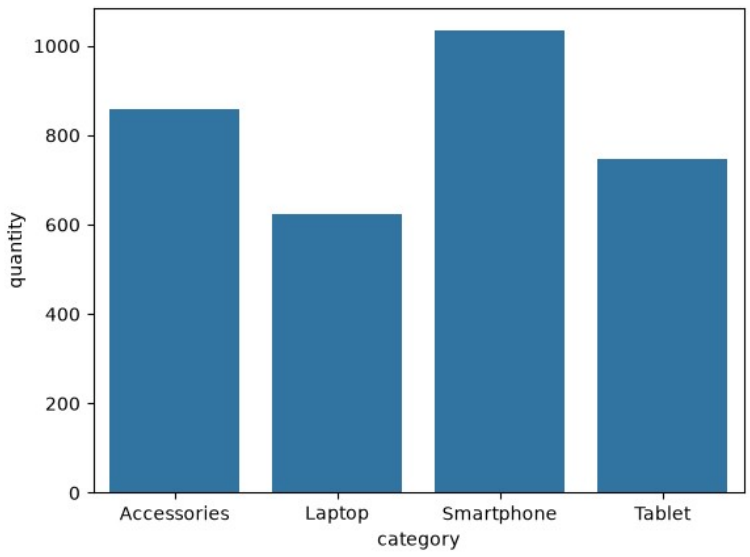


Figure 2. Units sold per category (seaborn).

5.2 Payment methods

Of the **958** orders, **Cash on Delivery** is the most used method (**283 orders**), followed closely by **Credit Card (277)** and **UPI (274)**. The top three are within about 3% of each other, so the ranking between them is not strong. **Debit Card** is far behind at **124**. Orders with no recorded payment method were removed.

Payment method	Orders	Share of 958 orders
Cash on Delivery	**283**	29.5%
Credit Card	277	28.9%
UPI	274	28.6%
Debit Card	124	12.9%

Table 6. Orders by payment method.

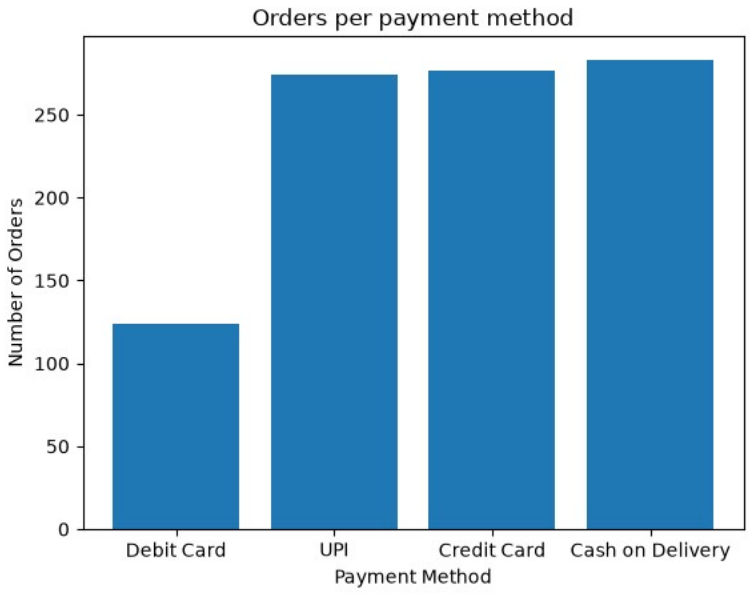


Figure 3. Number of orders per payment method.

5.3 Revenue by year

2024 generated **₹104.83M** (72.2%) against **₹40.31M** (27.8%) in **2025**. The 2025 figure covers only 1 January to 15 May, so the two years are not directly comparable.

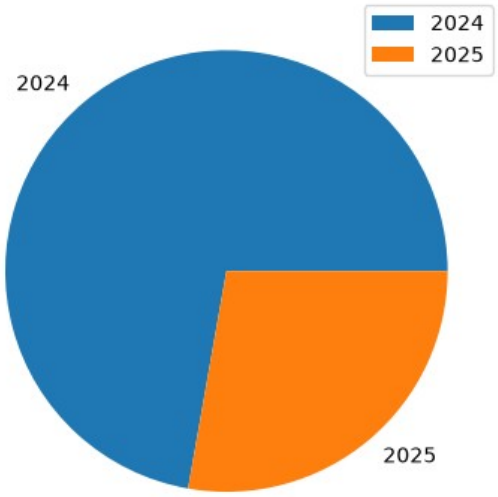


Figure 4. Share of revenue by year (2024 vs 2025).

5.4 Quarterly and monthly revenue

Q1 is the strongest quarter at **₹53.71M** and **Q3** the weakest at **₹23.25M**, with a small recovery in Q4 (₹25.15M). The monthly chart shows average revenue for each calendar month across the years it appears in. **April (₹10.55M)** and **January (₹10.14M)** are highest, while **November (₹6.02M)** is lowest.

Quarter	Revenue	Share of revenue
Q1	**₹53.71M**	37.0%
Q2	₹43.04M	29.7%
Q3	₹23.25M	16.0%
Q4	₹25.15M	17.3%

Table 7. Revenue by quarter.



Figure 5. Quarterly revenue.

Month	Average revenue	Share of revenue
April	**₹10.55M**	10.4%
January	₹10.14M	10.0%
October	₹9.95M	9.8%
March	₹9.30M	9.2%
June	₹9.29M	9.2%
December	₹9.18M	9.0%
September	₹8.41M	8.3%
July	₹8.12M	8.0%
February	₹7.42M	7.3%
August	₹6.72M	6.6%
May	₹6.32M	6.2%
November	₹6.02M	5.9%

Table 8. Average revenue by calendar month (share is of the sum of monthly averages).

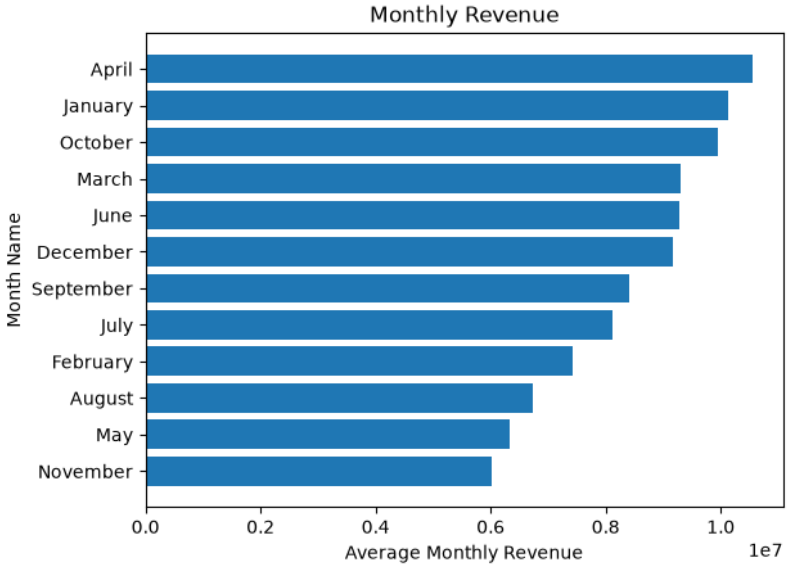


Figure 6. Average monthly revenue (highest at the top).

5.5 Revenue by city

Delhi is the top city with **₹44.41M** — about **30.6%** of revenue — and well ahead of **Mumbai (₹26.19M)**. **Hyderabad** (₹21.41M) and Bangalore (₹19.03M) follow, and **Chennai** is lowest at **₹9.82M**. Customers with no city were removed, so there is no Unknown group.

City	Revenue	Share of revenue
Delhi	**₹44.41M**	30.6%
Mumbai	₹26.19M	18.0%
Hyderabad	₹21.41M	14.8%
Bangalore	₹19.03M	13.1%
Pune	₹13.29M	9.2%
Kolkata	₹10.99M	7.6%
Chennai	₹9.82M	6.8%

Table 9. Revenue by city.

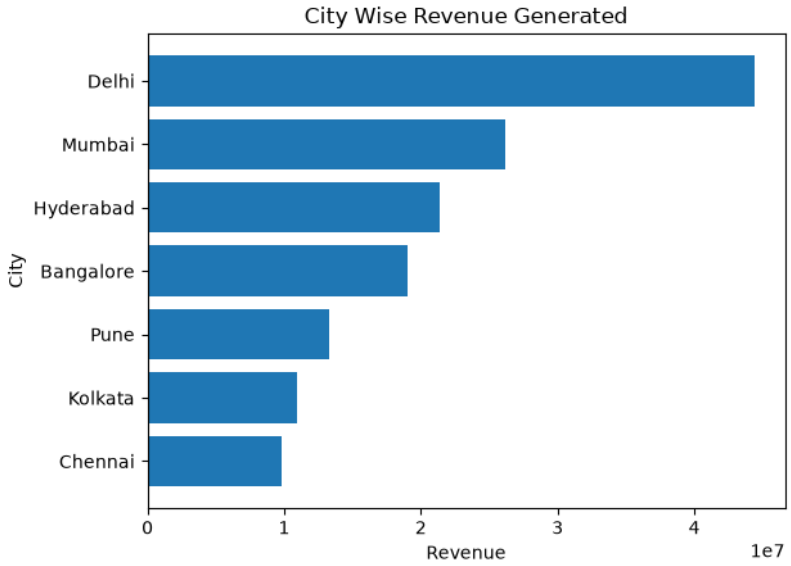


Figure 7. Revenue by city.

6. Assumptions & Limitations

- **Rows removed rather than filled.** Customers with no city and orders with no payment method were removed, so results describe **958 of 1,200** unique orders and **1,624 of 2,000** order items. Including the removed rows could change rankings that are close, such as payment methods.
- **Cash treated as Cash on Delivery.** The raw data has both "Cash" and "Cash on Delivery"; they were merged into one method, which is an assumption.
- **Price imputation. 1,011 of 2,050** unit prices (about **49%**) were filled with a median, and **57 of 90** base prices with the mean. This flattens price differences and can affect revenue by product and category. base_price is not used in any revenue calculation.
- **Uneven time coverage.** Orders span **1 January 2024 to 15 May 2025** (about 17 months), so January–May appear twice and June–December once. The monthly figures average each month across the years it appears in, and May is pulled down because May 2025 is only half-covered.
- **Unreliable product data.** Product names and categories do not always agree (for example, "Wireless Mouse" listed as a Laptop), so category-level results should be treated with care.
- **Missing names. 28** customer names are missing and set to Unknown; names are not used in the analysis.